## 3.2 Multiplication of Algebraic Expressions

While doing the product of algebraic expressions, we should follow the steps given below.

**Step 1:** Multiply the signs of the terms. That is, the product of two like signs are positive and the product of two unlike signs are negative.

| | | |
|---|---|---|
| Like signs | $(+)\times(+)=+$ | $(-)\times(-)=+$ |
| Unlike signs | $(+)\times(-)=-$ | $(-)\times(+)=-$ |

**Step 2:** Multiply the corresponding co-efficients of the terms.

**Step 3:** Multiply the variable factors by using laws of exponents.

If '$x$' is a variable and m, n are positive integers then,

$$x^m \times x^n = x^{m+n}$$

For example, $x^3 \times x^4 = x^{3+4} = x^7$

**Think**

Every algebraic expression is a polynomial. Is this statement true? Why?

**DO YOU KNOW?**

A polynomial is a special kind of algebraic expression. The difference between an algebraic expression and a polynomial is,

**Algebraic Expression**

May contains whole numbers, fractions, negative numbers as the power of their variables.

**Example:** $4x^{3/2} - 3x + 9$

$2y^2 + \frac{5}{y} - 3$, $3x^2 - 4x + 1$

**Polynomial**

contains only whole numbers as the power of their variables.

**Example:** $4x^2 - 3x + 9$

$2y^6 + 5y^3 - 3$

**Note**

Product of two terms is represented by the symbols ( ), dot (.) or $\times$.

For example,

multiplying $4x^2$ and $xy$ can be written in any one of the following ways.

| | | | | |
|---|---|---|---|---|
| $(4x^2)(xy)$ | $4x^2 \times xy$ | $4x^2(xy)$ | $(4x^2)\times xy$ | $4x^2 \cdot xy$ |

### 3.2.1 Multiplication of two or more monomials

Consider that, Geetha buys 3 pens each @ ₹5, how much she has to pay to the shopkeeper?



Geetha has to pay to the shopkeeper $= 3 \times ₹5$

$= ₹15$



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